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numpy.logspace() in Python

The `numpy.logspace()` function returns number spaces evenly w.r.t interval on a log scale.
Syntax :

```
numpy.logspace(start,
               stop,
               num = 50,
               endpoint = True,
               base = 10.0,
               dtype = None)
```

Parameters :

- > **start** : [float] start(base ** start) of interval range.
- > **stop** : [float] end(base ** stop) of interval range
- > **endpoint** : [boolean, optional]If True, stop is the last sample. By default, True
- > **num** : [int, optional] No. of samples to generate
- > **base** : [float, optional] Base of log scale. By default, equals 10.0
- > **dtype** : type of output array

Return :

-> ndarray

Code 1 : Explaining the use of logspace()

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Python

```
# Python Programming illustrating
# numpy.logspace method

import numpy as geek
```

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```
# base = 10, dtype = int
print("\n", geek.logspace(2.0, 3.0, num=5, dtype = int))
```

Output :

```
B
[ 121.          220.36039471  401.31159963  730.8527479  1331.          ]
B
[ 100.          177.827941   316.22776602  562.34132519 1000.          ]
B
[ 100  177  316  562 1000]
```

Code 2 : Graphical Representation of numpy.logspace() using matplotlib module – pylab

Python

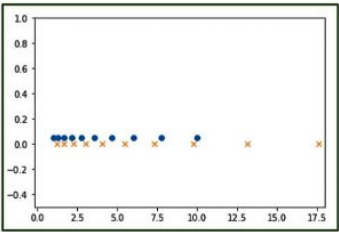
```
# Graphical Representation of numpy.logspace()
import numpy as geek
import pylab as p

# Start = 0
# End = 2
# Samples to generate = 10
x1 = geek.logspace(0, 1, 10)
y1 = geek.zeros(10)

# Start = 0.1
# End = 1.5
# Samples to generate = 12
x2 = geek.logspace(0.1, 1.5, 12)
y2 = geek.zeros(12)

p.plot(x1, y1+0.05, 'o')
p.xlim(-0.2, 18)
p.ylim(-0.5, 1)
p.plot(x2, y2, 'x')
```

Output :



Note :
These NumPy-Python programs won't run on online IDE's, so run them on your systems to explore them

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